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B.Sc. in Computer Science and Engineering
Practice Problem-2: 12 Sept, 2023

Subject: CSE 1202: Digital Logic Design Lab

Time: X

Full Marks: X

INSTRUCTIONS:

- a. Use + line of the breadboard for shorting **5V** and – line for shorting **GND**.
- b. Use a minimal number of **ICs** and **wires** for the hardware implementation.
- c. **Follow the orientation of the higher significant bit to lower significant bit from left to right or up to down.**

Write your **Group No** in the following box

Member	Roll	Name	Marks	
			Q1	Q2
1.				
2.				
3.				
4.				
5.				

Question – 1

Marks

Design a combinational circuit that takes 4-bit binary number as input and produces high output for the following cases:

$$\Sigma m(5,6,7,9,10,13,15)$$

Simplify the function using K-map and implement the circuit using Proteus. Label the inputs and output properly and connect the inputs with a 7-segment display.

X

Question – 2

Consider two binary strings $A[A_1 A_0]$ and $B[B_1 B_0]$. Design a combinational circuit in Proteus that produces the arithmetic sum of A and B and displays both inputs and output in 7-segment displays. Note that, both the inputs and outputs should be labeled properly.

X